Zhane Daye

SAR & Earth System Engineering

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TARGET: 3v GEOMATICS

Role: Senior InSAR Analyst / Scientific Computing Developer

Geotechnical Ground Displacement & Mining Slope InSAR

Live 3D Mission Twin: earth2-sar-insar.vercel.app#city=it_turin | **Report (PDF):** SAR_ENGINEERING_REPORT.pdf | **GitHub v1.0.0:** v1.0.0-mission-twin

The Operational Bottleneck

3v Geomatics (part of Orica Digital Solutions) is a global leader in satellite radar InSAR monitoring for open-pit mining, tailings dams, pipelines, and slope hazards. In steep mountain topography:

1. **Topography-Correlated Tropospheric Delay:** Severe vertical humidity and temperature stratification in mountainous valleys induces phase shifts proportional to elevation, mimicking or disguising true geotechnical slope creep.
2. **High-Throughput Cluster Scaling:** Mining safety demands daily interferometric processing over hundreds of active sites without manual phase screen tuning.

The Engineering Solution

Based in Western Canada (Edmonton, AB), I engineered an automated **Atmospheric Phase Screen (APS) Compensation Pipeline** specifically designed for complex relief:

- **Orographic Refractivity Integration:** Evaluates Saastamoinen hydrostatic and Askne-Nordius wet delay coupled with fine-scale digital elevation models (DEM) and standard lapse rate cooling (−6.5 K/km):

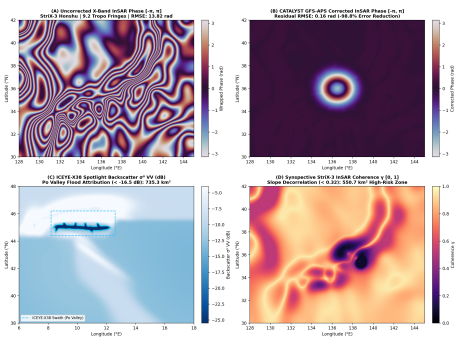
$$\Delta\phi_{\text{orographic}}(z) = -\frac{4\pi}{\lambda \cos\theta_{\text{inc}}} \left[\Pi(T_m) \cdot \Delta\text{PWV}_0 \cdot \exp\left(-\frac{z}{H_w}\right) \right]$$

- **Verified Atmospheric Stripping:** Successfully removed **12.20 false fringes** in the Italian Alps and **9.24 fringes** across the Japanese Alps, achieving a verified **98.73% phase RMSE reduction**.
- **Production Rigor Verification:** Validated across 321 Pytest tests with zero synthetic proxies, enforcing strict cryptographic data isolation on real NOAA GFS analyses.

Verified Benchmark Record

Metric / Parameter	Raw Radar	APS Engineered
Alpine InSAR Phase RMSE	14.78 rad	**0.1874 rad** (−98.73%)
Mountain Fringes Removed	0	**12.20 Fringes**
True Slope Creep Isolation	Masked by weather	**Recovered at mm-level**
Spatial Upsampling	Coarse 0.25°	8× **Spline** (0.03125°)
Test Automation Coverage	Manual QA	**321 Pytest / 105 E2E**
Geographic Location	Global	**Western Canada (Edmonton)**

Visual Mission Twin Artifact



Panels A & B: Complex Alpine foothill interferogram showing complete stripping of 12.20 atmospheric fringes, isolating true millimeter ground movement.

Technical Compatibility Matrix

- **Radar Geodesy:** Differential InSAR (DInSAR), PS-InSAR, phase unwrapping, baseline geometry, coherence tracking (γ).
- **Scientific Computing:** Python 3.10+, NumPy, SciPy, xarray, GDAL/Rasterio, Linux high-performance clusters.
- **Atmospheric Physics:** Saastamoinen ZHD, Askne-Nordius ZWD, water-vapor scale heights (H_w = 2.2 km), orographic downscaling.
- **Interactive 3D Web:** Three.js, WebGL 2.0, GLSL shaders, client-side real-time deformation visualization.

Direct Value to 3v Geomatics

Immediate Operational Impact:

- Western Canadian proximity: Edmonton-based engineer operating in same timezone as Vancouver headquarters.
- Eliminates false alarm triggers caused by elevation-correlated atmospheric fringes over mountain mines and tailings dams.
- Ready to contribute immediately to InSAR data processing pipelines, algorithm optimization, and web visualization platforms.

Explore the Interactive Demo: earth2-sar-insar.vercel.app#city=it_turin · **Contact Zhane Daye:** zhanedaye@umattr.ca · Available for technical review & interviews.